

# Coding & Solution

|               |                                            |
|---------------|--------------------------------------------|
| Date          | 15 July 2026                               |
| Team ID       | XXXXXX                                     |
| Project Name  | FloodGuard - AI/ML Flood Prediction System |
| Maximum Marks | 5 Marks                                    |

## Solution Summary

| Field                         | Details                                                                                                                                         |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository Link / URL         | <a href="https://github.com/BSG8710/flood-prediction-system">https://github.com/BSG8710/flood-prediction-system</a>                             |
| Programming Language(s)       | Python 3.12, JavaScript, HTML5, CSS3                                                                                                            |
| Framework(s) Used             | Flask 3.1, Flask-SQLAlchemy, scikit-learn, XGBoost                                                                                              |
| Key Features Implemented      | Model training/comparison pipeline; POST /api/predict endpoint; dynamic result rendering; prediction history; auto-generated EDA/insight charts |
| Pending / Incomplete Features | User authentication / login system (schema supports it; UI not yet built); PostgreSQL migration for persistent history on redeploy              |
| Setup / Run Instructions      | 1) pip install -r requirements.txt 2) python train.py 3) python app.py → visit <a href="http://localhost:5000">http://localhost:5000</a>        |

## Code Quality Checklist

| S.No | Criteria                                               | Status (Yes/No) |
|------|--------------------------------------------------------|-----------------|
| 1    | Code is modular and organized into functions / classes | Yes             |
| 2    | Meaningful variable and function names are used        | Yes             |
| 3    | Code includes comments / documentation where necessary | Yes             |
| 4    | Error handling is implemented for critical operations  | Yes             |
| 5    | The application runs without critical errors           | Yes             |
| 6    | Code is committed to a version control repository      | Yes             |

## Additional Notes / Comments

Deployed live on Render at: <https://floodguard-ytqr.onrender.com>